

Overview

This replication package contains code and data that reproduce the analyses in manuscript AER-2020-1283 “*Rebel on the Canal: Disrupted Trade Access and Social Conflict in China, 1650–1911*”. The replicator can run the master file in Stata to generate the data for the 11 figures and 14 tables in the paper and the online appendix. The code runs for about 1 hour. The package also provides the production files for the 11 maps in the paper and the online appendix.

Data Availability and Provenance Statements

Statement about Rights

We certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Summary of Availability

All data have been made publicly available.

Details on Each Data Source

Rebellions Data on rebellions between 1650 and 1911 were collected by the authors from *Qing Shilu* (The Veritable Records of Qing Emperors). The *Chinese Text Project* (Sturgeon, 2019) provides digitized versions of these books, available at <https://ctext.org>. The procedure of data collection and coding is described in Online Appendix C. The dataset contains the number of documented events at the county-year level, for 575 counties between 1650 and 1911.

Data file:

`Data/Raw/rawrebellion.dta`

CHGIS Shapefiles for 1820 county and other geographic features can be downloaded from CHGIS, Version 4 (Harvard Yenching Institution and Fudan Center for Historical Geography, 2007) at <http://dx.doi.org/10.7910/DVN/PDGOZ0> upon agreeing to the CHGIS Terms of use (Berman, 2017). The shapefile data were then processed in ArcMap to create the geographic variables used in the paper. The details of the ArcMap pre-processing that was carried out is provided in the Appendix of this README.

Data files:

Data/Raw/Geo_raw.xlsx
Data/Raw/V4wgs84/samplemap.dbf
Data/Raw/V4wgs84/samplemap.shp
Data/Raw/Canal/canal.shp
Data/Raw/Canal/canal.dbf
Data/Raw/CanalProj/intertab.txt
Data/Raw/coast.xlsx
Data/Raw/ToHuang.xlsx
Data/Raw/ToYangtze.xlsx
Data/Raw/nearcourier.txt
Data/Raw/oldyellowriver.xls
Data/Raw/towns/town1820.shp
Data/Raw/towns/town1820.dbf
Data/Raw/towns/town1911.shp
Data/Raw/towns/town1911.dbf
Data/Raw/Chong/chong.shp
Data/Raw/Chong/chong.dbf
Data/Raw/countyadmin/countyadmin.shp
Data/Raw/countyadmin/countyadmin.dbf
Data/Raw/Canal/admintocanal.txt
Data/Raw/Canal/admintocourier.txt
Data/Raw/Canal/chongtocanal.txt
Data/Raw/Canal/chongtocourier.txt
Data/Raw/Canal/town1820tocanal.txt
Data/Raw/Canal/town1820tocourier.txt
Data/Raw/Canal/town1911tocanal.txt
Data/Raw/Canal/town1911tocourier.txt
Data/Raw/lengths/Ming_Routes_2016_Identity_St1_TableToExcel.xls
Data/Raw/lengths/chgis_1820_county_seat_TableToExcel.xls
Data/Raw/lengths/Changjiang_Output_Identity_S_TableToExcel.xls
Data/Raw/lengths/v4_1911_cnty_pgn_gbk_TableToExcel.xls

Initial Population Data on county level population in 1546 are digitized by the authors from Liang (1980).

Data file:

Data/Raw/countypop_ming.xls

USGS GTOPO30 Grid-cell level terrain elevation data can be downloaded from Global 30 Arc-Second Elevation (US Geological Survey, 1997) at <https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-global-30-arc-second-elevation-gtopo30> upon agreeing to the USGS Terms of use. The raster data were then processed in ArcMap to be associated with the 1820 county shapefiles contained in CHGIS, which generates terrain elevation and ruggedness index variables used in the paper. The details of the ArcMap pre-processing that was carried out is provided in the Appendix of this README.

Data files:

Data/Raw/eleraster/elev1.txt

...

Data/Raw/eleraster/elev1204.txt

Taiping Rebellion Data on regions affected by the Taiping Rebellion are extracted by the authors from the map published in Guo (1989).

Data file:

Data/Raw/taipingregion.xlsx

Imperial Soldiers Data on imperial soldiers are digitized by the authors from Luo (1984).

Data File:

Data/Raw/Soldier_all.xlsx

GAEZ v3 Grid-cell level crop suitability data can be downloaded from IIASA and FAO (2012) at <https://www.fao.org/nr/gaez/en/> upon agreeing to the FAO Terms of use (Fischer et al., 2012). The data were accessed on March 16, 2016. The raster data were then processed in ArcMap to be associated with the 1820 county shapefiles contained in CHGIS, which generates the suitability index variables used in the paper. The details of the ArcMap pre-processing that was carried out is provided in the Appendix of this README.

Data files:

Data/Raw/suitability/wlrice.txt

Data/Raw/suitability/joinwheat

Climate Anomaly Grid-cell level historical climate anomaly data from Mann et al. (2009) were obtained from <https://www.science.org/doi/10.1126/science.1177303> and saved in csv format.

Data files:

Data/Raw/Mann/allproxyfieldrecon.csv

Data/Raw/Mann/longlat.csv

Data on other historical variables Additional historical variables used in the paper were digitized by Chen and Kung (2016) from various archival sources: i) data on the adoption of new world crops were digitized from prefectural level local gazetteers (difang zhi); ii) data on historical droughts and floods were digitized from Chinese Academy of Meteorological Sciences (1981); iii) data on grain prices were computed based on information from Wang (2008) and Institute of Economics (2010); data on historical population were digitized from Cao (2001). See Chen and Kung (2016) for details of the data construction process. The crosswalk to merge the dataset is manually created by the authors.

Data files:

Data/Raw/maize_raw_data.dta

Data/Raw/climate500.dta

Data/Raw/grainprice.dta

Data/Raw/PrefwID.xlsx

Shipping amount The amount of tribute rice shipped via the canal are digitized by the authors from Li and Jiang (2008), pages 38–42.

Data file:

Data/Raw/amount.dta

Green Gang Senior Members The number of Green Gang Senior Members is digitized by the authors from Zhang (1941).

Data file:

Data/Raw/Addvar2.xlsx

Replication Instructions

Datasets

The raw data in the Data/Raw folder are listed and summarized below:

- Data/Raw/rawrebellion.dta (provided) is collected from *Qing Shilu* (The Veritable Records of Qing Emperors)
- The following data files (all provided) are generated in ArcMap based on data from CHGISv4:
 - Data/Raw/V4wgs84/samplemap.dbf
 - Data/Raw/V4wgs84/samplemap.shp
 - Data/Raw/Geo_raw.xlsx
 - Data/Raw/Canal/canal.shp
 - Data/Raw/Canal/canal.dbf
 - Data/Raw/coast.xlsx
 - Data/Raw/ToHuang.xlsx
 - Data/Raw/ToYangtze.xlsx
 - Data/Raw/nearcourier.txt
 - Data/Raw/oldyellowriver.xls
 - Data/Raw/towns/town1820.shp
 - Data/Raw/towns/town1820.dbf
 - Data/Raw/towns/town1911.shp
 - Data/Raw/towns/town1911.dbf
 - Data/Raw/Chong/chong.shp
 - Data/Raw/Chong/chong.dbf
 - Data/Raw/countyadmin/countyadmin.shp
 - Data/Raw/countyadmin/countyadmin.dbf
 - Data/Raw/Canal/admintocanal.txt
 - Data/Raw/Canal/admintocourier.txt
 - Data/Raw/Canal/chongtocanal.txt
 - Data/Raw/Canal/chongtocourier.txt
 - Data/Raw/Canal/town1820tocanal.txt

- Data/Raw/Canal/town1820tocourier.txt
- Data/Raw/Canal/town1911tocanal.txt
- Data/Raw/Canal/town1911tocourier.txt
- Data/Raw/CanalProj/intertab.txt
- Data/Raw/lengths/Ming_Routes_2016_Identity_St1_TableToExcel.xls
- Data/Raw/lengths/chgis_1820_county_seat_TableToExcel.xls
- Data/Raw/lengths/Changjiang_Output_Identity_S_TableToExcel.xls
- Data/Raw/lengths/v4_1911_cnty_pgn_gbk_TableToExcel.xls
- Data/Raw/countypop_ming.xls (provided) is collected from Liang (1980).
- Data files in the Data/Raw/eleraster (provided) folder are generated in ArcMap based on data from GTOPO30.
- Data/Raw/taipingregion.xlsx (provided) is collected from Guo (1989).
- Data/Raw/Soldier_all.xlsx (provided) is collected from Luo (1984).
- Data/Raw/suitability/wlrice.txt and Data/Raw/suitability/joinwheat.shp (provided) are generated in ArcMap based on data from GAEZv3.
- Data/Raw/Mann/allproxyfielldrecon.csv and Data/Raw/Mann/longlat.csv (provided) are from Mann et al.(2009) and have been converted from Matlab to csv format.
- Data/Raw/maize_raw_data.dta, Data/Raw/climate500.dta, and Data/Raw/grainprice.dta (provided) are from Chen and Kung (2016). Data/Raw/PrefwID.xlsx (provided) provides the crosswalk to link the data.
- Data/Raw/amount.dta (provided) is collected from Li and Jiang (2008).
- Data/Raw/Addvar2.xlsx (provided) is collected from Zhang (1941).

Programs

The folder `Program` contains all programs to replicate the 11 figures and 14 tables in the main body of the paper and its online appendix.

- `Program/master.do` runs all programs to prepare the data and to replicate the figures and tables.
- `Program/setup.do` installs all required Stata packages required for reproducing the analysis.

- The folder `Program/Adofile` contains the Stata packages used in the analysis that are not available at SSC. These files need to be copied to the Stata `ado` directory on the local machine (this step can be completed automatically by running `Program/setup.do`).
 - `Program/Adofile/ols_spatial_HAC.ado` is provided by Hsiang (2010), available at <http://www.fight-entropy.com/2010/06/standard-error-adjustment-ols-for.html>
- The folder `Program/clean` contains programs that process the raw data in the (`Data/Raw` folder, which generate final dataset `Data/Final/rebellion.dta`) for analysis.
- The folder `Program/Analysis` contains programs that reproduces the figures and tables in the paper and its online appendix. Each program identifies the table or figure it creates (e.g., `Program/Analysis/table2.do`). Output files are called appropriate names (`Results/Tables/table2.tex`, `Results/Figures/figure1.pdf`, etc.)

Requirements

- Stata (Code was last run with version 17, minimum with 16)
 - `shp2dta`
 - `cic`
 - `coefplot`
 - `dm0014`
 - `erepost`
 - `estout`
 - `ftools`
 - `geodist`
 - `geoinpoly`
 - `hdfe`
 - `labutil`
 - `matmap`
 - `moremata`
 - `mvfiles`
 - `ols_spatial_HAC`
 - `reghdfe`

- synth
- synth_runner
- st0500
- ArcMap (Map was last displayed with version 10.6)
- System support for UTF-8 encoding is required
- Approximate time needed to reproduce the analyses on a laptop with intel(R) Core(TM) i7-10750H CPU @ 2.60GHz 5.00Ghz is: 1 hour

Replication Procedure

Instructions to Replicator

1. Edit `Program/master.do`, line 12 to adjust the default path
2. Comment out line 15 and 22 unless running on a new machine for the first time
3. Run `Program/master.do` to run all steps in sequence
4. (optional) Compile `Results/overview.tex` to view all replicated results in one place.

Details

`Program/master.do` calls the following programs in sequence

- `Program/setup.do`, unless commented out, installs the packages required for the replication, and set up the environment.
- `Program/Clean/clean.do` imports each of the raw data in `Data/Raw` folder and produces the final dataset `Data/Final/rebellion.dta` for analysis. It calls the following programs in the process:
 - `Program/Clean/clean_ruggedness.do`
 - `Program/Clean/clean_mann.do`
- Programs in the `Program/Analysis` folder produces Figure 2, Figure 4 and Table 1–7 in the main body of the paper, and Figure A1–A7, Figure C6b, Figure C8b and Table A1–A7 of the online appendix.

Maps

Several figures in the paper are presented as Maps. The production files of these maps are provided in **Program/Maps** folder, which can be viewed in ArcMap. The names of the files correspond to the numbering of figures in the paper.

- Programs in the **Program/Maps** folder produces Figure 1, Figure 3 in the main body of the paper, and Figure B1–B3, Figure C3–C5, Figure C6a, Figure C7–C8a of the online appendix. The results are saved in the **Results** folder.

References

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Yue, Zumou, G.William Skinner, and Mark Henderson, Davis: University of California, Regional Systems Analysis Project, Jan 2007.

Zhang, Boying, 1941, *Encyclopedia of the Green Gang (Qingbang Tongcao Huihai, in Chinese)*,

Appendix: Details of Arcmap Preprocessing

CHGIS

- Data/Raw/V4wgs84/samplemap.shp and the associated .dbf file
 1. Import the shapefile v4_1911_cnty_pgn_gbk.shp from CHGIS, Version 4 (2007) to ArcMap as a layer
 2. Click on the “Select” button from the main task bar and then “Select by Attributes”
 3. Select features where:
LEV1_PY IN ('Anhui', 'Henan', 'Jiangsu', 'Shangdong', 'Zhejiang', 'Zhili')
 4. Right click on the layer, click on “Selection” and then “Create layer from selected features”
 5. Rename the new layer as samplecounties
 6. Click on the “ArcToolBox” button from the Tool bar
 7. Double click on the “Conversion Tools” → “To Shapefile” → “Feature Class To Shapefile (multiple)” tool from the “ArcToolBox” window
 8. Select the samplecounties layer and export it as Data/Raw/V4wgs84/samplemap.shp (the tool will generate the associated .dbf file as well)
- Data/Raw/Canal/canal.shp and the associated .dbf file
 1. Import the shapefile v4_1820_rvr_lin_gbk.shp from CHGIS, Version 4 (2007) to ArcMap as a layer
 2. Follow previous steps to open “Select by Attributes” and select features where:
RIVERS_ID IN (2107, 2369 2775)
 3. Follow previous steps to create a new layer and rename it as canal_main
 4. Follow previous steps to create a new layer named as canal_extended where the selection function is:
RIVERS_ID IN (47, 1184, 2107, 2303, 2369, 2397, 2414, 2775)
 5. Click on the “ArcToolBox” button from the Tool bar
 6. Double click on the “Conversion Tools” → “To Shapefile” → “Feature Class To Shapefile (multiple)” tool from the “ArcToolBox” window
 7. Select the samplecounties layer and export it as Data/Raw/V4wgs84/samplemap.shp (the tool will generate the associated .dbf file as well)

8. Use the “Feature Class To Shapefile (multiple)” tool to export the `canal_main` layer as `Data/Raw/Canal/canal.shp` (and the associated `.dbf` file)
- `Data/Raw/Geo_raw.xlsx`
 1. Follow previous steps to create the `samplecounties` and the `canal_extended` layers in ArcMap
 2. Double click on the “Analysis Tools” → “Proximity” → “Near” tool from the “ArcToolBox” window
 3. Select `samplecounties` as the input feature and `canal_extended` as the near feature, check the “Angle” box and execute the tool
 4. Use the “Conversion Tools” → “Excel” → “Table To Excel” tool to export to `Data/Raw/Geo_raw.xlsx`
 - `Data/Raw/CanalProj/intertab.txt`
 1. Follow previous steps to create the `samplecounties` and the `canal_main` layers in ArcMap
 2. Double click the “Analysis Tools” → “Statistics” → “Tabulate Intersection” tool from the “ArcToolBox” window
 3. Select `samplecounties` as the Input Zone Features, `OBJECTID` as the Zone Fields, `canal_main` as the Input Class Features, and `OBJECTID` as the Class Fields, and execute the tool to generate the intersection table
 4. Right click on the intersection table, select “Data” → to export the table as `Data/Raw/CanalProj/intertab.txt` (setting “Text File” as the Save as Type)
 - `Data/Raw/coast.xlsx`
 1. Follow previous steps to create the `samplecounties` layer in ArcMap
 2. Import the shapefile `v4_1820_coast_lin.shp` from CHGIS, Version 4 (2007) to ArcMap as a layer
 3. Use the “Near” tool to calculate the distance from `samplecounties` to this layer
 4. Use the “Table to Excel” tool to export the data as `Data/Raw/coast.xlsx`
 - `Data/Raw/ToHuang.xlsx`
 1. Follow previous steps to create the `samplecounties` layer in ArcMap
 2. Import the `v4_1820_rvr_lin_gbk.shp` shapefile to ArcMap as a layer

3. Follow previous steps to create a new layer where the selection function is: `RIVERS_ID = 2776`
 4. Use the “Near” tool to calculate the distance from `samplecounties` to this layer
 5. Use the “Table to Excel” tool to export the result as `Data/Raw/ToHuang.xlsx`
- `Data/Raw/ToYangtze.xlsx`
 1. Follow previous steps to create the `samplecounties` layer in ArcMap
 2. Import the `v4_1820_rvr_lin_gbk.shp` shapefile to ArcMap as a layer
 3. Follow previous steps to create a new layer named as `yangtze` where the selection function is: `RIVERS_ID = 2778`
 4. Use the “Near” tool to calculate the distance from `samplecounties` to this layer
 5. Use the “Table to Excel” tool to export the result as `Data/Raw/ToYangtze.xlsx`
 - `Data/Raw/nearcourier.txt`
 1. Follow previous steps to create the `samplecounties` layer in ArcMap
 2. Import the `Courier_Routes` shapefile from Harvard WorldMap (feature name: “Courier_Routes”) ¹ to ArcMap as a layer
 3. Use the “Near” (or “Generate Near Table”) tool to calculate the distance from `samplecounties` to this layer
 4. Export the result as `Data/Raw/nearcourier.txt`
 - `Data/Raw/oldyellowriver.xls`
 1. Follow previous steps to create the `samplecounties` layer in ArcMap
 2. Import the `Yellow_River_1368_1855_0` shapefile from Harvard WorldMap (feature name: “Yellow River 1368-1855”) ² to ArcMap as a layer
 3. Use the “Near” tool to calculate the distance from `samplecounties` to this layer
 4. Use the “Table to Excel” tool to export the data as `Data/Raw/oldyellowriver.xls`

¹<https://worldmap.maps.arcgis.com/home/item.html?id=e1da2b19eb9c4c09853995cf64ce5928>, last accessed Feb 18, 2022

²<https://worldmap.maps.arcgis.com/home/item.html?id=d153bb2ae389426e99ff452d0552ff63c>, last accessed Feb 18, 2022.

- `Data/Raw/towns/town1820.shp` (and the associated `.dbf` file) is taken directly from `v4_1820_twn_pts_gbk.shp`
- `Data/Raw/towns/town1911.shp` (and the associated `.dbf` file) is taken directly from `v4_1911_twn_pts_gbk.shp`
- `Data/Raw/Chong/chong.shp` (and the associated `.dbf` file) is taken directly from the `China_1820_Busy_Xian_0.shp` shapefile from Harvard WorldMap (feature name: “China 1820 Busy Xian”)³
- `Data/Raw/countyadmin/countyadmin.shp` (and the associated `.dbf` file) is taken directly from the `v4_1911_cnty_pts_gbk.shp` shapefile from CHGIS, Version 4 (2007)
- `Data/Raw/Canal/admintocanal.txt`
 1. Follow previous steps to create the `canal_main` layer in ArcMap
 2. Import the `v4_1911_cnty_pts_gbk.shp` shapefile from CHGIS, Version 4 (2007) to ArcMap as a layer
 3. Use the “Near” (or “Generate Near Table”) tool to calculate the distance between the two layers
 4. Export the result as `Data/Raw/Canal/admintocanal.txt`
- `Data/Raw/Canal/admintocourier.txt`
 1. Import the `v4_1911_cnty_pts_gbk.shp` shapefile from CHGIS, Version 4 (2007) to ArcMap as a layer
 2. Import the `Courier_Routes` shapefile from Harvard WorldMap (feature name: “Courier_Routes”) to ArcMap as a layer
 3. Use the “Near” (or “Generate Near Table”) tool to calculate the distance between the two layers
 4. Export the result as `Data/Raw/Canal/admintocourier.txt`
- `Data/Raw/Canal/chongtocanal.txt`
 1. Follow previous steps to create the `canal_main` layer in ArcMap
 2. Import the `China_1820_Busy_Xian_0.shp` shapefile from Harvard WorldMap (feature name: “China 1820 Busy Xian”) to ArcMap as a layer
 3. Use the “Near” (or “Generate Near Table”) tool to calculate the distance between the two layers
 4. Export the result as `Data/Raw/Canal/chongtocanal.txt`
- `Data/Raw/Canal/chongtocourier.txt`

³<https://worldmap.maps.arcgis.com/home/item.html?id=0dcb7de695e646659cbff68f9fb82c43>, last accessed Feb 18, 2022.

1. Import the `Courier_Routes` shapefile from Harvard WorldMap (feature name: “Courier_Routes”) to ArcMap as a layer
 2. Import the `China_1820_Busy_Xian_0.shp` shapefile from Harvard WorldMap (feature name: “China 1820 Busy Xian”) to ArcMap as a layer
 3. Use the “Near” (or “Generate Near Table”) tool to calculate the distance between the two layers
 4. Export the result as `Data/Raw/Canal/chongtocourier.txt`
- `Data/Raw/Canal/town1820tocanal.txt`
 1. Follow previous steps to create the `canal_main` layer in ArcMap
 2. Import the `v4_1820_twn_pts_gbk.shp` shapefile from CHGIS, Version 4 (2007) to ArcMap as a layer
 3. Use the “Near” (or “Generate Near Table”) tool to calculate the distance between the two layers
 4. Export the result as `Data/Raw/Canal/town1820tocanal.txt`
 - `Data/Raw/Canal/town1820tocourier.txt`
 1. Import the `Courier_Routes` shapefile from Harvard WorldMap (feature name: “Courier_Routes”) to ArcMap as a layer
 2. Import the `v4_1820_twn_pts_gbk.shp` shapefile from CHGIS, Version 4 (2007) to ArcMap as a layer
 3. Use the “Near” (or “Generate Near Table”) tool to calculate the distance between the two layers
 4. Export the result as `Data/Raw/Canal/town1820tocourier.txt`
 - `Data/Raw/Canal/town1911tocanal.txt`
 1. Follow previous steps to create the `canal_main` layer in ArcMap
 2. Import the `v4_1911_twn_pts_gbk.shp` shapefile from CHGIS, Version 4 (2007) to ArcMap as a layer
 3. Use the “Near” (or “Generate Near Table”) tool to calculate the distance between the two layers
 4. Export the result as `Data/Raw/Canal/town1911tocanal.txt`
 - `Data/Raw/Canal/town1911tocourier.txt`
 1. Import the `Courier_Routes` shapefile from Harvard WorldMap (feature name: “Courier_Routes”) to ArcMap as a layer
 2. Import the `v4_1911_twn_pts_gbk.shp` shapefile from CHGIS, Version 4 (2007) to ArcMap as a layer

3. Use the “Near” (or “Generate Near Table”) tool to calculate the distance between the two layers
 4. Export the result as `Data/Raw/Canal/town1911tocourier.txt`
- `Data/Raw/lengths/Ming_Routes_2016_Identity_St1_TableToExcel.xls`
 1. Follow previous steps to create the `samplecounties` layer in ArcMap
 2. Import the `Ming_Routes_2016.shp` shapefile from CHGIS, Version 6 (2017) to ArcMap as a layer
 3. Open the “Analysis” → “Overlay” → “Identity” tool from the ArcTool-Box, choose the new layer as the input feature and `samplecounties` as the identity feature and execute the tool
 4. Right click on the layer generated in the previous step and click on “Open Attribute Table”
 5. Click on “Table Options” from the attribute table and then “Add Field”
 6. Set “length” as the field name and select “double” as the Type
 7. Right click on the “length” field from the attribute table and click on “Calculate Geometry”
 8. Select “Length” as Property and execute the tool
 9. Open the “Analysis Tools” → “Statistics” → “Summary Statistics” tool from the ArcToolBox
 10. Select the current layer as Input Table, “length” as Statistics Field(s) and “SUM” as the Statistic Type, and execute the tool
 11. Use the “Conversion Tools” → “Excel” → “Table to Excel” tool from the ArcToolBox to export the table generated from the previous step as `Data/Raw/lengths/Ming_Routes_2016_Identity_St1_TableToExcel.xls`
 - `chgis_1820_county_seat_TableToExcel.xls`
 1. Follow previous steps to create the `yangtze` layer in ArcMap
 2. Import the `v4_1820_cnty_pts_gbk.shp` shapefile to ArcMap as a layer
 3. Use the “Near” tool to calculate the distance between the two layers
 4. Use the “Table to Excel” tool to export the result as `chgis_1820_county_seat_TableToExcel.xls`
 - `Changjiang_Output_Identity_S_TableToExcel.xls`
 1. Follow previous steps to create the `samplecounties` layer in ArcMap

2. Follow previous steps to create the **yangtze** layer in ArcMap
 3. Open the “Identity” tool, set the **yangtze** layer as the Input Feature and the **samplecounties** layer as the Identity Feature, and execute the tool
 4. Add a new field to the newly generated layer, and use the “Calculate Geometry” tool to calculate the “Length” property
 5. Open the “Summary Statistics” tool, set “length” as Statistics Field(s) and “SUM” as the Statistic Type, and execute the tool
 6. Use the “Table to Excel” tool to export the result as **Changjiang_Output_Identity_S_TableToExcel.xls**
- **v4_1911_cnty_pgn_gbk_TableToExcel.xls**
 1. Import the **v4_1911_cnty_pgn_gbk.shp** shapefile to ArcMap as a layer
 2. Use the “Table to Excel” tool to export it as **v4_1911_cnty_pgn_gbk_TableToExcel.xls**

USGS GTOPO30

1. Follow previous steps to create the **samplecounties** layer in ArcMap
2. Import the USGS GTOPO30 raster file to ArcMap as a layer named as **elevation**
3. Open the “ArcToolBox” Window, right click anywhere within the window, select “Add Toolbox”, click on the “New Toolbox” icon to create a new toolbox, and click “Open”
4. Right click on the newly created toolbox and select “New” → “Model”
5. Right click on the newly created model and select “Edit” to open the model builder window
6. Build a model to iterate over each county polygon in the **samplecounties** layer by inserting the “Feature Selection” iterator (select “OBJECT” as the Group by Fields parameter), and perform the following operations within each iteration (see Figure 1 for an illustration of the pipeline)
 - (a) Execute the “Data Management” → “Raster” → “Raster Processing” → “Clip” tool, where the Input Raster is set as **elevation** and the Output Extent is set as the selected feature from the “Feature Selection” iterator
 - (b) Execute the “Raster to ASCII” tool, where the Input Raster is set as the output from the “Clip” tool, and the Output ASCII file is set as **Data/Raw/eleraster/elev\%Value\%.TXT**

- (c) Execute the “Delete” tool, where the Input Data Element parameter is the output from the “Clip” tool
- 7. Click on the “Run” icon from the model builder window to execute the entire model, which generates files in the **Data/Raw/eleraster** folder

GAEZ v3

- **Data/Raw/suitability/wlrice.txt**
 1. Follow previous steps to create the **samplecounties** layer in ArcMap
 2. Import the raster file for wetland rice suitability (under irrigation & medium input) from GAEZ v3 as a layer in Arcmap
 3. Open the “Data Management” → “Raster” → “Raster Processing” → “Clip” tool from the ArcToolBox, set the imported raster layer as the Input Raster and **samplecounties** as the Output Extent, and execute the tool
 4. Use the “Raster to ASCII” to export the clipped raster layer as **Data/Raw/suitability/wlrice.txt**
- **Data/Raw/suitability/joinwheat.shp** (and the associated .dbf file)
 1. Follow previous steps to create the **samplecounties** layer in ArcMap
 2. Import the raster file for wetland rice suitability (under irrigation & medium input) from GAEZ v3 as a layer in Arcmap
 3. Open the “Conversion Tools” → “From Raster” → “Raster to Point” tool from the ArcToolBox, select the imported raster as the Input Raster and “Value” as the Field, and execute the tool
 4. Right click on the **samplecounties** layer from the Table of Contents window and select “Joins and Relates” → “Join”
 5. Select “Join data from another layer based on spatial location” as the task and the points converted from the raster file as the layer to join
 6. Select “Each polygon will be given a summary of the numeric attributes of points that fall inside it, and a count field showing how many points inside it”, and check “Average”
 7. Execute the spatial join operation
 8. Use the “Feature Class to Shapefile” tool to export the result as **Data/Raw/suitability/wlrice.txt**

Figure 1: Model builder for preprocessing elevation data

